Kafka message contract

This document describes the messages exchanged between the UI and indicators tools on the CEDAR Kafka cluster.

Flow

1. The UI publishes a **request** to the pilot's request topic.
2. The tools runs the requested indicators on the requested entities.
3. The tools publishe a single **response** to the pilot's results topic.
4. The UI matches the response to its request using `requestID`.

Pilot	Request topic	Results topic
Italian	`it_analyse`	`it_results_available`
Slovenian	`sl_analyse`	`sl_results_available`

The request topics are shared, so other tools may also answer the same `requestID` on the results topic. Use `toolName` or `indicator_id` to tell the responses apart.

!NOTE!: Knoledge Graph has "id": "ten_b0835bc998" and "ID_Tender": "B0835BC998". We use **ID_Tender** in all entities.

Request (UI → tool)

```
{
  "issuer": "ui",
  "requestID": "5f1c2e0a-8a3b-4c1d-9f2e-7b6a1d3c4e5f",
  "entities": [
    { "entityType": "tender", "entityId": "B0835BC998" }
  ],
  "indicators": ["CERTH_LLM_UC6_AI2", "CERTH_LLM_UC2_AI3"],
  "kg": { "versionId": "v1.1" },
  "accounting": {},
  "status": {}
}
```

Field	Type	Required	Notes
`issuer`	string	yes	Who sent the request.

`requestID`	string	yes	Unique ID; it is copied into the response.
`entities[]`	object[]	yes	Entities to analyse.
`entities[].entityType`	string	yes	`"tender"` or `"bid"`. The indicators currently process only `"tender"`.
`entities[].entityId`	string	yes	ID as it appears in the pilot's knowledge graph.
`indicators[]`	string[]	no	Plain `indicator_id` strings. An empty or missing list runs **all** indicators for the pilot.
`kg.versionId`	string	yes	Knowledge graph version; it is copied into the response.
`accounting`	object	no	Reserved.
`status`	object	no	Reserved.

Requests that fail validation are logged and skipped, and no response is sent. A common cause is sending `indicators` as objects (`[{"indicatorId": "..."}]`) instead of strings.

Response (tool → UI)

```
{
  "requestID": "5f1c2e0a-8a3b-4c1d-9f2e-7b6a1d3c4e5f",
  "responseID": "ae45f99d-247a-4a0d-806e-b8014ef7d5db",
  "toolName": "CERTH_LLM",
  "analysis": {
    "startTime": 1789635170592,
    "endTime": 1789635187821,
    "indicators": [
      {
        "id": "8e85e664-49a1-4260-b6cf-ae0ae3c34854",
        "indicator_id": "CERTH_LLM_UC6_AI2",
        "entityID": "B0835BC998",
        "risk_level": 1.0,
        "additional_info": [
          {
            "id": "ceb961bc-ddfc-4ccb-80d2-948d01b77e02",
            "title": "Price Deviation from Market Analysis (Updated
            "value": "Bid 1 is the main issue: it is underpriced by
            "type": "string",
            "risk_level": 1.0,
            "description": "Analysis of how bid prices in this tend
            "range": { "from": 0.0, "to": 1.0 }
          }
        ]
      }
    ]
  },
  "source": {
    "kg": {
      "versionId": "v1.1",
      "entities": [
        { "entityType": "tender", "entityId": "B0835BC998" }
      ]
    }
  }
}
```


Field	Type	Always present	Notes
requestID	string	yes	Same as the request.
responseID	string	yes	Unique ID for this response.
			example

toolName	string	no	"CERTH_LLM"
analysis.startTime	integer	yes	Epoch milliseconds: when the request was published.
analysis.endTime	integer	yes	Epoch milliseconds: when the response was built.
analysis.indicators[]	object[]	yes	One item per indicator per entity.
analysis.indicators[].id	string	yes	Unique ID for this result.
analysis.indicators[].indicator_id	string	yes	See **Indicators map excel**.
analysis.indicators[].entityID	string	yes	The entity this result is about.
analysis.indicators[].risk_level	number	yes	Overall risk, **0.0–1.0**.
analysis.indicators[].additional_info[]	object[]	yes	Details to display
additional_info[].id	string	yes	Unique ID.
additional_info[].title	string	yes	Heading to display.
additional_info[].type	string	yes	Type of value (numeric, text, long_text, url, list_tenders, list_bids, image.)

`additional_info[].value`	any	no	Main result
`additional_info[].units`	string	no	For example `"percentage"`.
`additional_info[].description`	string	no	Method and supporting data (multi-line text).
`additional_info[].risk_level`	number	no	Risk for this detail.
`additional_info[].range`	object	no	`{ "from": number, "to": number }`: the scale of the value or risk. (I think we have agreed on `0-1`)
`source.kg.versionId`	string	yes	Same as the request.
`source.kg.entities[]`	object[]	yes	Every entity used in the analysis, including similar tenders used for comparison. Same shape as the request's `entities[]`.

Rules for consumers

- **Optional fields are omitted, never `null`.** Check whether a key exists before reading it.
- **One response per request.** Results for all requested indicators and entities are combined into one message.
- **No response when nothing was produced.** If none of the requested indicators return a result (for example, only unknown IDs were requested), nothing is published. Use a timeout on the UI side.
- **Failed analyses are still published.** They have `"value": "Analysis Failed"`, `risk_level` `0.0`, and the reason in `description`. Check `value` before treating

`0.0` as "no risk".